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SUPERSEDING
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FEDERAL SPECIFICATION

GLOVES, X-RAY PROTECTIVE, LEATHER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a left-hand and a right-hand, lead-impregnated, X-ray protective glove, suitable for protecting the hands and forearms of the wearer from exposure to X-ray during fluoroscopic X-ray examinations.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP–B–585—Boxes, Wood, Wirebound.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Boxes, Fiber.
PPP–B–665—Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–T–7807—Thread, Nylon.
MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.**—Material shall conform in all respects to the requirements of this specification.

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3.1.1 *Lead-impregnated leather.* — The lead-impregnated leather shall be high-quality split leather of uniform thickness of approximately 2½ ounces, as measured by a Woburn or similar standard accepted leather gage. The lead equivalent of each thickness of lead-impregnated leather shall be approximately 0.20 millimeter (mm.) at 100 peak kilovolts (PKV).

3.1.2 *Leather cover.* — The leather cover shall be approximately 1½ ounces, as measured by a Woburn or a similar accepted standard leather gage, and shall be soft and pliable, high-grade, smooth-grained calf, or equal, free from defects such as holes, tears, and thin patch areas.

3.1.3 *Liner.*—The liner shall be approximately 2½ ounces, as measured by a Woburn or similar standard accepted leather gage, and shall be durable, soft, split pigskin leather, impregnated on the unexposed side with a lead equivalent of approximately 0.10 millimeters lead.

3.1.4 *Thread.*—Thread shall be nylon conforming to Military Specification MIL–T–7807. The outside stitching shall be type I, size E (minimum); inner stitching shall be 3-ply, type I, size B (minimum).

3.2 **Design.**—Protective gloves shall be of the gauntlet style, of the conventional five-finger design with sewed-in leather lining, and shall provide full coverage of the hands and wrists. Gloves shall be designed and manufactured to afford a maximum degree of comfort to the wearer.

3.3 **Dimensions.**—Each glove shall have an overall length, from the tip of the middle finger to the extremity of the cuff, of not less than 14 inches. The overall width of each glove, across the distal metacarpal plane, shall be 5¼ inches (plus or minus ⅛ inch), thereby accommodating a hand measurement of size 9. The length measurements of the thumb and fingers shall be as follows (plus or minus ⅛ inch):

(*a*) Thumb (as measured at junction with base of index finger), 3⅛-inches.
(*b*) Index finger (as measured at junction of index finger and middle fingers), 3-inches.
(*c*) Middle finger (as measured at junction of index and middle fingers), 3½-inches.
(*d*) Annular finger (as measured at junction of middle finger and annular finger), 3⅜-inches.
(*e*) Little finger (as measured at junction of annular and small finger), 2¾-inches.

3.4 **Leather and lead impregnation.**—The body of the glove shall be of soft, pliable leather impregnated with lead compounds. The protective impregnation shall be uniform throughout. The gloves shall afford the wearer protection equivalent to no less than 0.5 mm. of pure lead at 100 PKV. Each glove shall consist of three component inner gloves in addition to the leather covering. The innermost glove shall be the liner (see 3.1.3). The other two inner gloves shall each be impregnated with lead on their top and bottom surfaces of approximately 0.10 mm. of lead for each surface, and a total of 0.20 mm. of lead for each individual inner glove (see 3.1.1). The lead equivalent of the complete glove shall be not less than 1.0 mm. of lead when tested as specified in 4.4.1.

3.5 **Leather covering.**—The leather covering shall be permanently and smoothly attached to the outside of the glove. All edges shall either be turned and sewed, or shall be bound with binding strips of the same leather and sewed.

3.6 **Liner.**—The liner shall be an insert of soft leather of such durability as to assure long wear.

3.7 **Seams and stitching.**—All seams and stitching shall be in conformity with the best commercial glove manufacturing practice, and shall be sealed or otherwise covered in such a manner as to afford protection against

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X-rays equivalent to no less than 0.5 mm. of pure lead at 100 PKV.

3.8 **Flexibility and texture.**—The finished gloves, including seams and stitch lines, shall be soft and sufficiently flexible to permit manipulation of the X-ray machine and table controls. The surface texture of both the insert and external leather covering, including seams and stitches, shall be smooth throughout.

3.9 **Identification marking.** — Each glove shall have a label permanently affixed to the lining, legibly marked with the name or registered trade-mark of the manufacturer, stock number, contract number, the lead equivalent of the material used and the voltage for which this lead equivalent was determined, and the letters "U.S.". Size shall be as required for ready readability and identification.

3.10 **Workmanship.** — Workmanship shall be first class throughout. All seams shall afford no less protection against X-radiation than the base material, and the construction shall be such as not to impair flexibility. Gloves shall be without defects, such as unsealed stitch holes and cracks, and shall be opaque throughout to X-radiation to the degree herein specified.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The provisions contained in this section are applicable to inspection performed by and for the Government. Sampling, inspection, and test procedures for components or materials described shall be in accordance with this specification and with specifications and standards referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.1.1 *Supplier responsibilities for inspection.*—Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and shall be made available to the Government, upon the Government's request, at any time or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.2 Inspection, as used in this specification, is defined as both examination (such as, visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and the test results obtained, and indicating that material is as specified, may be furnished.

4.2 **Sampling.** — Sampling shall be conducted in accordance with Military Standard MIL–STD–105 and Appendix thereto.

	Inspection level	AQL (percent defective)
For visual examination:		
Major	II	1.0
Total	II	4.0
For dimensional examination	L5	2.5
For nondestructive component examination	L5	*
For end item testing	III	*

* Acceptance number shall be zero for all sample sizes.

4.2.1 *Components.*—Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the gloves to perform tests as required by the

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applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above. No test result shall fall below the minimum or exceed the maximum values specified.

4.2.2 Where possible, the same sample shall be used for the determination of two or more tests.

4.3 **Inspection.**—The gloves shall be inspected to determine compliance with all requirements contained in sections 3 and 5 of this specification. Nonconformance with these requirements will be permitted to the extent indicated in 4.2.

4.3.1 *Classification of defects.*—Inspection shall be conducted in accordance with the following classifications of defects:

	Categories and defects**
Major:	
101.	The liner, 2 inner gloves, and cover stitched together at points other than the cuff.
102.	Leather covering not free of holes, tears, or thin patch areas.
103.	Stitching in all component parts not secure and free from breaks (not suitably repaired).
104.	Glove components not securely assembled and free of creases and undue ridges.
105.	Glove not soft and sufficiently flexible to permit operation of X-ray machine controls.
106.	Specified label to be affixed to glove liner not present, complete, or correct.
Minor:	
201.	Liner and external leather covering not smooth.
202.	Label or printing on unit package not legible and free from defacing marks, scratches, or tears.
203.	Gloves not packaged in the flat position (unfolded).

** Inspection is not restricted to the classified possible defects listed above.

4.3.2 *Dimensional examination.* — The gloves shall be inspected for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests.

4.4.1 *X-ray tests.*—The samples shall be radiographed by placing the glove, or gloves, alongside a suitable lead penetrometer graduated in steps of 0.1 mm. of pure sheet lead covering. The radiographs shall be examined by means of a suitable densitometer to determine compliance with this specification. The following technical factors are suggested for use in carrying out the tests:

100 Kilovolts
10 Milliamperes
60 Seconds
30 Inches distance
Standard X-ray films.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing protection used are defined in Federal Standard No. 102.

5.1 **Packaging.**—Packaging shall be level A or C, as specified (see 6.1):

5.1.1 *Unit of issue.*—One glove, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Federal stock number	Unit package	Intermediate package	Exterior container
8415–603–0011 or 8415–603–0012	1 unit	Not required	24 units

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5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, PPP–B–665, or PPP–B–676. Glove shall not be folded. Glove shall be powdered or otherwise treated to prevent binding or tacking. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**—Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specifications PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *For civil agencies.*—Unit packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *For military agencies.*

5.3.2.1 *Unit package.*—Each unit package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.2 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that 6.2(c), 6.2(d), and 6.5 shall not be applicable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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(*a*) Title number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

8415–603–0011 GLOVE, X-RAY PROTECTIVE, Leather, Left Hand.

8415–603–0012 GLOVE, X-RAY PROTECTIVE, Leather, Right Hand.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 **Transportation description.**—Transportation description and minimum weight applicable to this commodity are:

Rail:

Gloves, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Gloves, not otherwise indexed.
Motor volume minimum weight 12,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M Q
Navy—Md
Air Force.

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